

Portfolio Management (5. Portfolio Construction)

Alphas and Portfolio Construction

- ▶ A typical portfolio construction process is a constrained optimization problem that involves an **objective function** (mean-variance utility function) and **constraints** (business mandates such as no short position, limiting risk/exposure etc....). The solution to this problem is the **asset allocation**.

$$\begin{array}{ll}
 \text{Objective} & \max_w \mathcal{U}(\mathbf{w}) \\
 \text{s.t.} & g(\mathbf{w}) \leq \mathbf{c}
 \end{array}$$

Handwritten red annotations: An arrow points from the word 'Objective' to the word 'max'. Another arrow points from the word 's.t.' to the word 'g'. A third arrow points from the word 'w' in the objective function to the word 'w' in the constraint function. The word 'w' in the constraint function is also underlined.

- ▶ *excess relative* In active portfolio management, we work in the active space i.e., instead of raw returns, we use active returns, instead of means(μ), we work with alphas(α) and instead of standard deviation of returns(σ) as risk, we use standard deviation of active returns(ω) aka active risk

Modified Alphas and Risk Aversion

- ▶ We can match the active holdings $\mathbf{h}_{PA}^*$ and active risk ω_P^* of any sophisticated portfolio construction process by modifying alphas and using appropriate risk aversion coefficient under an **unconstrained** mean/variance optimization problem
- ▶ The objective function of the optimization problem is given as

Objective

$$\max_{\mathbf{w}} \mathcal{U}(\mathbf{w}) = \underbrace{\mathbf{w}^T \boldsymbol{\alpha}'}_{\text{(modified alpha)}} - \lambda \underbrace{\mathbf{w}^T \boldsymbol{\Omega} \mathbf{w}}_{\text{(Cov matrix)}}$$

For max utility, setting first order derivative to 0, we get $\frac{\partial \mathcal{U}}{\partial \mathbf{w}} = \boldsymbol{\alpha}' - 2\lambda \boldsymbol{\Omega} \mathbf{w}^* = \mathbf{0} \Rightarrow \boldsymbol{\alpha}' = 2\lambda \boldsymbol{\Omega} \mathbf{w}^*$

To match the active holdings $\mathbf{h}_{PA}^*$ of the underlying portfolio construction process, alphas need to be modified such as

$$\boldsymbol{\alpha}' = 2\lambda \boldsymbol{\Omega} \mathbf{h}_{PA}^*$$

Information ratio (IR)

- ▶ The job of an active portfolio manager is to outperform a benchmark. He/she does it by trading alpha against active risk. The **information ratio** captures this trade-off. It's a **fundamental constant** that defines a manager. Think of this ratio as a measure of performance consistency

$$0.5 \rightarrow \frac{1}{1} \rightarrow IR_p = \frac{\alpha_p}{\omega_p}$$

Handwritten diagram showing the formula $IR_p = \frac{\alpha_p}{\omega_p}$ with red annotations. A red circle is around IR_p , and a red box is around ω_p . A red arrow points from the fraction to the number 0.5, and another red arrow points from the fraction to the number 1.

- ▶ A manager can deliver more alpha by taking more risk. However, the above formula holds exactly in an unconstrained optimization problem. We can re-write the utility function as

$$\rightarrow u(\omega_p) = \alpha_p IR_p \omega_p - \lambda \omega_p^2$$

Handwritten diagram showing the utility function $u(\omega_p) = \alpha_p IR_p \omega_p - \lambda \omega_p^2$ with red annotations. A red circle is around IR_p , and a red arrow points from α_p to the circle. The entire expression is underlined in red.

Solving first order condition for max utility, $\omega_p^* = \frac{IR_p}{2\lambda}$ $u^* = \frac{IR_p^2}{4\lambda}$

The modified alphas are given as

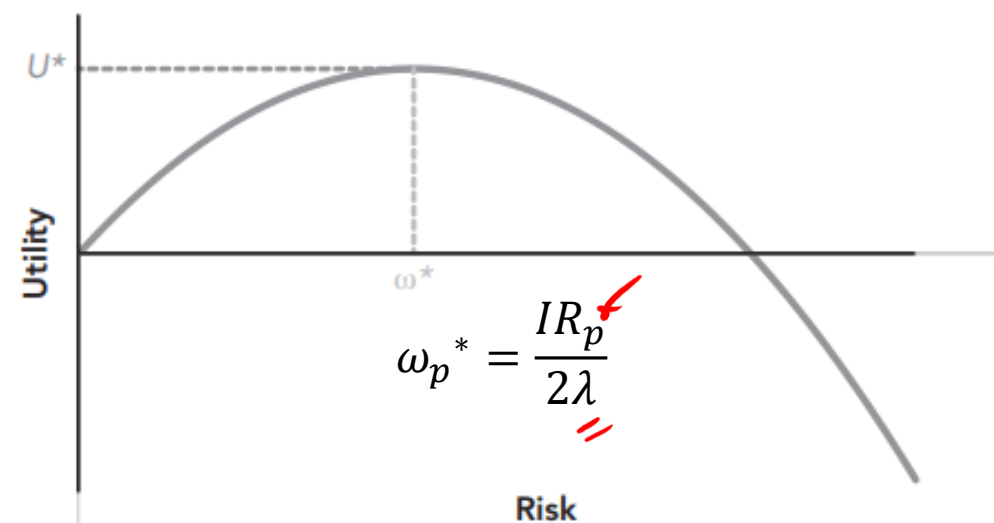
$$\alpha' = \left(\frac{IR_p}{\omega_p^*} \right) \Omega h_{PA}^*$$

Optimal Active Risk



- The optimal risk the investor will tolerate is **inversely proportional** to the risk aversion of the investor and **directly proportional** to the Information ratio of the manager.

$$U^* = \frac{IR_p^2}{4\lambda}$$



Signals to Alpha

- ▶ Active managers use their skill to turn signals into return forecasts and future alpha. Yes, alpha can be estimated from ex-post realized returns, but the real skill of manager is to forecast returns ex ante. Let's consider a simple linear model

$$r_n(t) = \beta_n r_B(t) + \theta_n(t)$$

(stock return)
(bmk return)
(residual return)

- ▶ Unconditional return forecast assumes $\mathbb{E}[\theta_n(t)] = 0$ which leads to benchmark holdings. For an active manager, the expected residual return conditional on some information g is not 0 but α . The signals can be earning forecasts, broker buy/sell recommendations etc....

$$\mathbb{E}[\theta_n(t) | g] = \alpha_n$$

(signal)

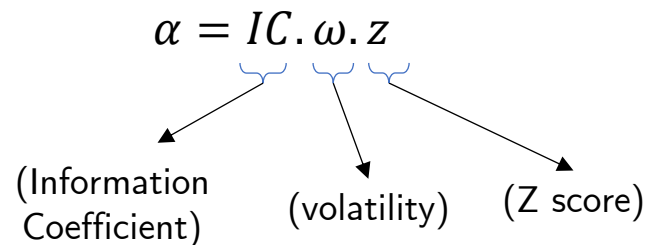
Anatomy of Alpha

- Alpha embeds skill, volatility and signal. The best linear unbiased estimate of residual return given a raw signal is

$$\mathbb{E}[\theta|g] = \mathbb{E}[\theta] + \frac{\text{Cov}(\theta, g)}{\text{Var}(g)} [g - \mathbb{E}[g]]$$

- It can be re-written as

$$\mathbb{E}[\theta|g] - \mathbb{E}[\theta] = \text{Corr}(\theta, g) \cdot \text{StDev}(\theta) \cdot \left[\frac{g - \mathbb{E}[g]}{\text{StDev}(g)} \right]$$

$$\alpha = IC \cdot \omega \cdot z$$


(Information Coefficient) (volatility) (Z score)

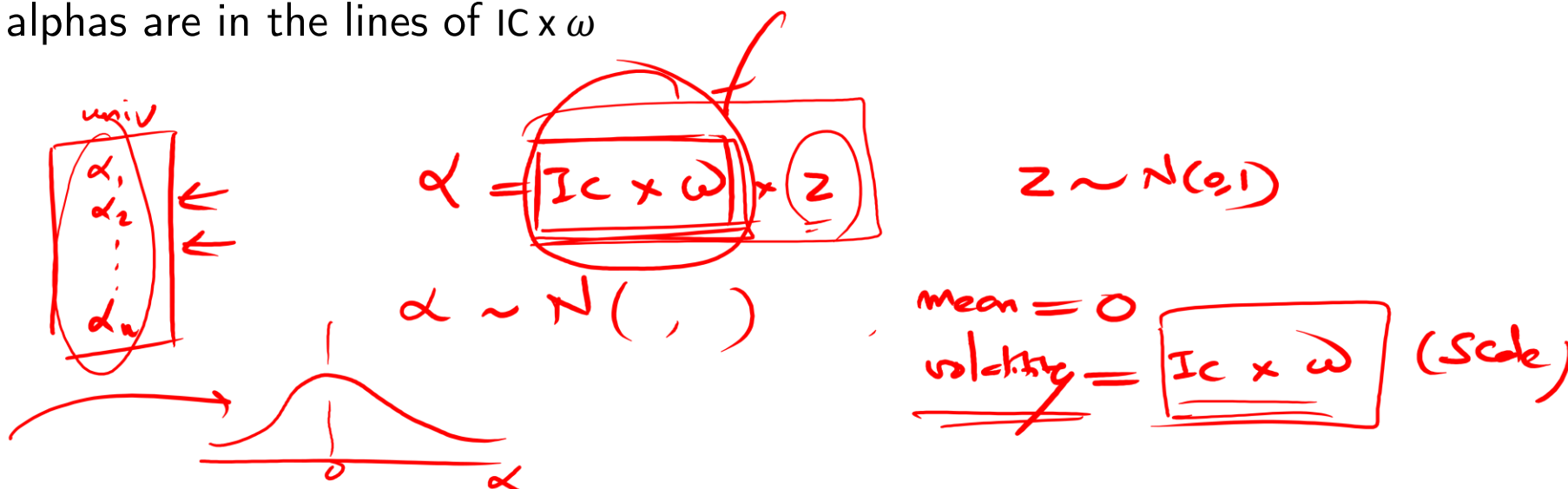
The above formula gives a very intuitive break up of alpha in terms of skill, residual risk and score. IC of 5% is considered good and 10% and beyond is considered great.

Refining Alpha – Scale

- ▶ The IC and volatility is roughly assumed to be a constant. The z-score is a standard normal random. So, alpha has a normal distribution with mean 0 and standard deviation or scale = $IC \times \omega$.

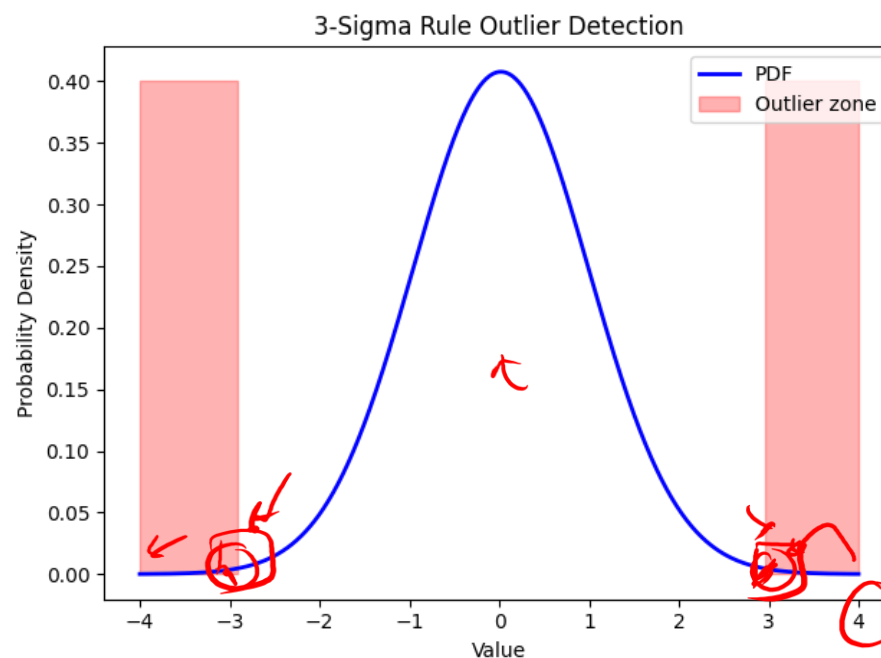
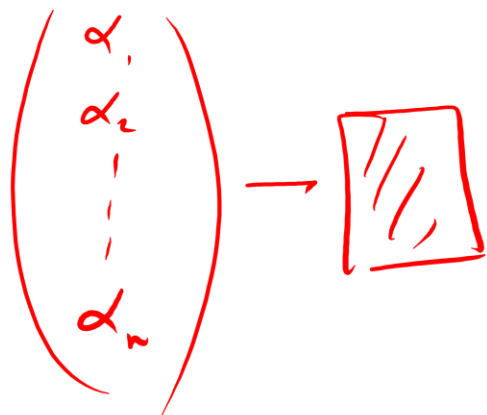
$$\sigma(\alpha) = IC \times \omega$$

- ▶ For example, an information coefficient (IC) of 5% and a residual risk of 30% leads to a scale of 1.5%.
- ▶ Scale of alpha depends on manager's information coefficient. So before entering optimization, if the alphas do not have proper scale, we need to rescale them to ensure the resulting standard deviation of alphas are in the lines of $IC \times \omega$



Refining Alpha – Trimming

- ▶ Alpha trimming refers to the traditional outlier treatment during data processing. Very large and small alphas can skew the optimization result. So, outliers need to be properly treated before they are fed into the optimization.
- ▶ The rule of thumb is to set $\alpha = 0$ for those which are beyond $\pm 3 \times \text{scale}$ if it's based on questionable data set. Else if the data is genuine, then include them by capping at the $\pm 3 \times \text{scale}$ level.



$$\alpha = 0$$

Refining Alpha – Neutralization

- ▶ We can remove biases and undesirable bets in alpha. We call this process **neutralization**.
- ▶ Benchmark neutralization refers to adjusting alphas, so the overall benchmark has 0 alpha
- ▶ In multi-factor models, one can choose to neutralize any factor
- ▶ The process to benchmark neutralization is as follows
 - Start with raw or modified alphas
 - Find the benchmark alpha as weighted average of the alphas
 - Adjust alpha of each security as $\alpha_i' \leftarrow \alpha_i - \beta_i \alpha_B$

$$\sum w_i \alpha_i = \alpha_B' \rightarrow 0$$

$$\alpha_i \leftarrow$$



Transaction costs

- ▶ The transaction cost is the cost of moving from one portfolio to another. Any buying/selling action attracts transaction cost.
- ▶ With only alpha and risk, all we need to do is adjust the weights which is a one-dimensional problem. But with transaction costs, it becomes a two-dimensional problem. On top of alpha and risk trade off one needs to consider alpha and TC trade offs.

Exercise 1 – Stock 1's current price is \$100. The price of stock 1 will increase to \$102 in the next 6 months and then remain at \$102. Stock 2's current price is also \$100. The price of stock 2 will increase to \$108 over the next 24 months and then remain at \$108. The cost of buying and selling each stock is \$0.75. The annual alpha for both stock 1 and stock 2 is 4 percent. Assume every 6 months, you can find a similar stock as stock 1. Find TC adjusted annual % return and TC for both stocks.

$$\{P_1\} \xrightarrow{TC} \{P_2\}$$

Different risk aversions

- ▶ When using mean variance utility framework, with multiple sources of risk, the risk aversion may not be the same. One may wish to use a higher risk aversion for specific risk than that of common factor risk.

$$\max_w \mathcal{U} = \alpha_p - [\lambda_{CF} \omega_{P,CF}^2 + \lambda_{SF} \omega_{P,SF}^2]$$

The equation is annotated with red markings: an upward arrow points to the max operator, a downward arrow points to the alpha_p term, and red circles and arrows highlight the lambda terms and their corresponding variance terms, indicating the focus on risk aversion parameters.

- ▶ Two reasons to consider implementing higher risk aversion for specific risk
 - One, since specific risk arises from large bets on specific assets, using a larger penalty will reduce the size of the bet on that specific asset
 - Two, for managers of multiple portfolios, aversion to specific risk will reduce dispersion. The portfolios will be pushed towards holding the same names.

Alpha coverage



- ▶ **Case 1** – if we forecast returns of stock which are not in the benchmark

This is easy to handle. Just pretend that the stock is included in the benchmark with 0 weight. Any such position in the portfolio will be an active position.

- ▶ **Case 2** – if we don't have forecasts for some stocks which are in the benchmark

Let's divide the benchmark set of stocks into two subsets. Set N_1 consists of stocks for which we have forecast and alpha. The remaining set N_0 consists of stocks for which we do not have any forecast. We need to do the following

- For all stocks in N_1 , adjust the alpha to $\alpha_i \leftarrow \alpha_i - \frac{\sum_{i \in N_1} \alpha_i w_i}{\sum_{i \in N_1} w_i}$
- For all the stocks in N_0 , set alpha = 0

This adjustment will lead to neutral alpha for benchmark.

Rebalancing Frequency

- ▶ In short term, noise dominates while in long term fundamental dominate. Rebalancing for short term horizon will involve frequent reactions to noise, not signal. But transaction costs, remain the same whether we react to noise or signal.
- ▶ With arrival of new information, whether a manager needs to trade or not can be based on comparing the **marginal value added** to the transaction cost. The marginal value added is given as

$$\text{MCVA}_n = \underbrace{\alpha_n}_{\substack{\text{(marginal contribution} \\ \text{to value added)}}} - 2 \cdot \lambda_A \cdot \omega_p \cdot \underbrace{\text{MCAR}_n}_{\substack{\text{(marginal contribution} \\ \text{to active risk)}}$$

- ▶ If PC is the purchase cost and SC is the sales cost, then manager should not trade a given stock if $-SC_n \leq \text{MCVA}_n \leq PC_n$. We can also put bounds around alpha for no trade condition

$$2 \cdot \lambda_A \cdot \omega_p \cdot \text{MCAR}_n - SC_n \leq \alpha_n \leq PC_n + 2 \cdot \lambda_A \cdot \omega_p \cdot \text{MCAR}_n \quad \checkmark$$

Portfolio Construction Techniques

- ▶ Portfolio optimization maximizes the value added less transaction costs

$$\alpha_p - \lambda_A \cdot \omega_p^2 - TC$$

▶ 1. Screens

- ❑ Rank the stocks by alpha
- ❑ Choose the first say 50 stocks. (put them into buy, hold, sell category)
- ❑ Equal-weight (or Mkt Cap weight) the stocks

\$50

buy	10
hold	5
sell	25

Features

- ❑ Simplicity, easy to code
- ❑ Robust (Not affected by noisy estimates of alpha as it's based on ranking)
- ❑ Ignores all other information of alpha apart from rankings
- ❑ Does not protect against biases



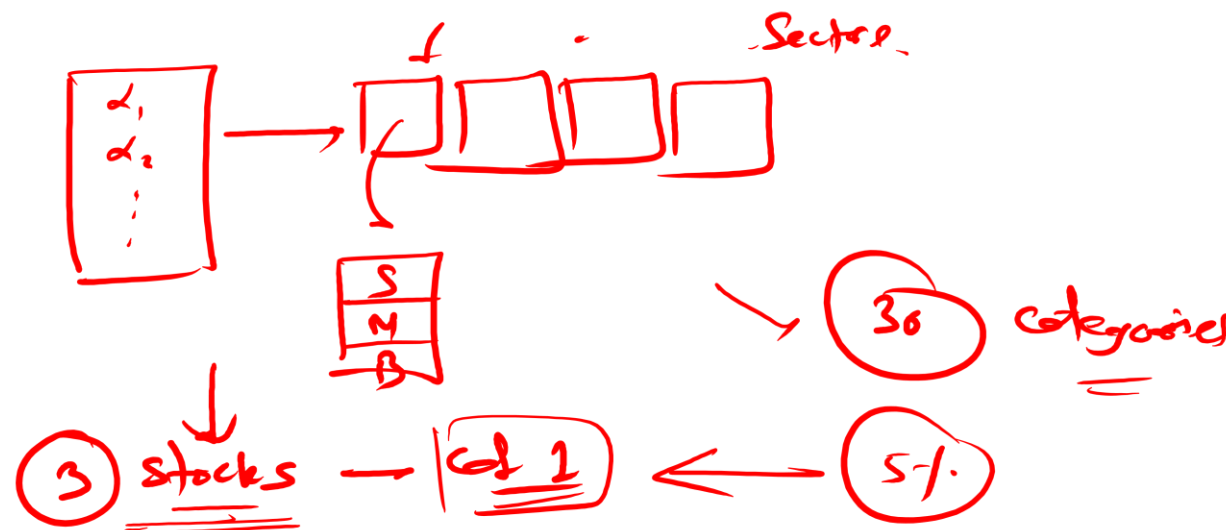
Portfolio Construction Techniques

► 2. **Stratification** (glorified screening)

- ❑ break down into sub-populations/categories (say 10 economic sectors, and each sector into 3 sizes; big, small, medium ~ 30 categories)
- ❑ Mimic the screening exercise in each category (rank the stocks based on alpha and put them into buy, sell, hold buckets to keep the turnover reasonable)
- ❑ Weight the stocks in a way that category weight in the portfolio matches the category weight in the benchmark

Features

- ❑ Similar features as screening
- ❑ Selection of categories has subjectivity



Portfolio Construction Techniques

▶ 3. Linear Programming (LP)

- ❑ A linear programming is characterized by a linear objective function and linear constraints
- ❑ Each stock can have multiple characteristics such as alpha, beta, size, industry etc. The goal of the linear programming is to maximize alpha less transaction costs while remaining close to the benchmark portfolio in the risk dimension.

$$\text{Max } \alpha_p = \alpha_1 w_1 + \alpha_2 w_2 \dots$$

$$20\% \leq w_1 + w_2 \dots w_n \leq 30\%$$

Features

- ❑ Easy to code and solution/interpretation is very tractable (Simplex)
- ❑ Hard to control features like cardinality (# of stocks)

$$0.9 \leq w_1 \beta_1 + w_2 \beta_2 + \dots \leq 1.1$$

Portfolio Construction Techniques

► 4. Quadratic Programming (QP)

- ❑ A Quadratic programming is all-encompassing – it's the ultimate portfolio engineering
- ❑ It's the most inclusive; alpha, risk and costs

Features

- ❑ The most comprehensive, state of the art portfolio optimization used
- ❑ Includes LP as a special case
- ❑ Too many inputs are needed (Cov matrix of 50 stocks) – risk of noisy estimates



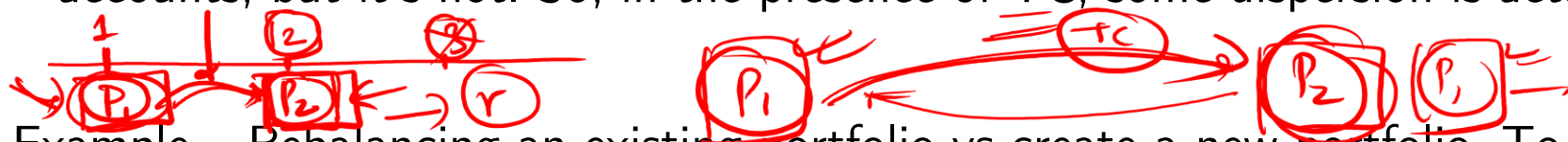
Dispersion



- ▶ Dispersion is a client support problem for a manager who manages multiple accounts. It's the difference between the maximum return and the minimum return of accounts under the same manager.
- ▶ If the holdings are the same across accounts, then dispersion will disappear. If all accounts see the same alpha, same benchmark, same investment process, then why the returns differ ? There are a few sources of dispersion

❑ Client driven. Client imposes various constraints which are beyond a manager's control

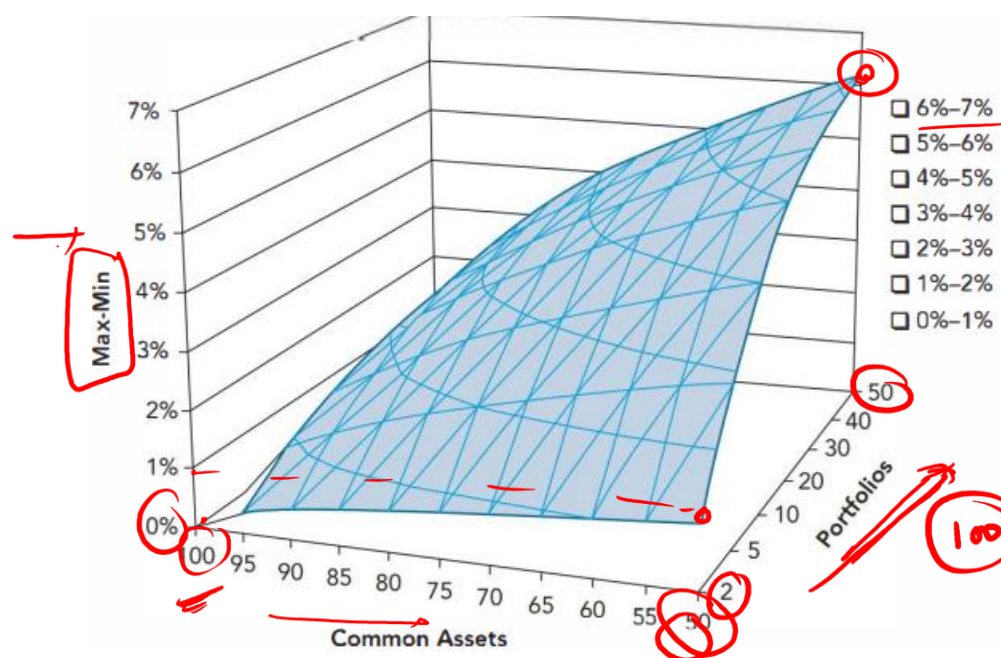
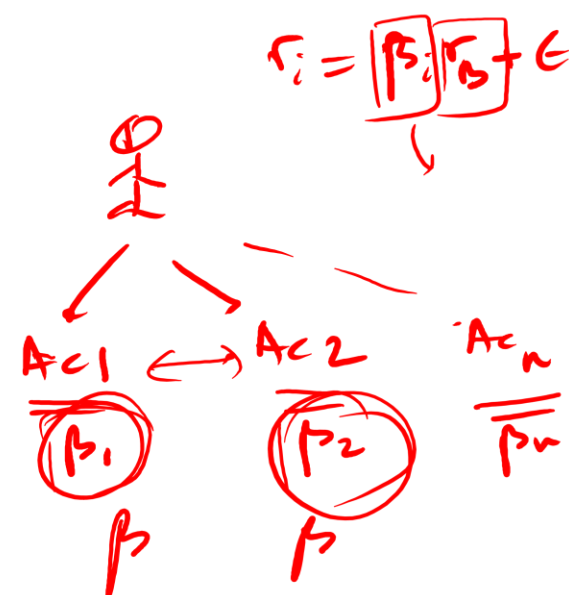
❑ History and Transaction cost – The portfolios were created at different points in time for various clients. The assets are different. Now, if $TC = 0$, then we could force the same asset allocation for all accounts, but it's not. So, in the presence of TC , some dispersion is actually optimal.



Example – Rebalancing an existing portfolio vs create a new portfolio. To reduce dispersion to 0, either bring the existing portfolio to the new portfolio and incur significant transaction costs OR allocate the new portfolio exactly as the existing portfolio and sacrifice returns. By accepting some dispersion, the manager achieves good composite returns on both portfolios combined.

Characterizing Dispersion

- ▶ First consider the difference in betas or factor exposures. Let's say the portfolios under a manager have betas ranging from 0.9 to 1.1. If the market return is 35% in a year, just because the difference in betas the dispersion would be $(1.1 - 0.9) \times 35\% = 7\%$
- ▶ Even if the manager manages to keep the betas same across portfolios, due to differences in assets across portfolios, there will be dispersion. For example, if we consider 50 different portfolios, each portfolio has 100 stocks with varying number of stocks they have in common, below is a diagram that illustrates dispersion



Managing Dispersion

- Does dispersion decrease with time ? If alphas and risk stay constant, then dispersion will never disappear. There will always be a transaction cost barrier. The remaining tracking error (between the initial portfolio and optimal portfolio) is bounded based on transaction cost and risk aversion

Handwritten diagram illustrating the relationship between portfolio P, transaction cost TC, and risk aversion λ . The diagram shows a box labeled P with an arrow pointing to a box labeled TC, which then points to a circle labeled λ . Below this, the formula $\omega^2 \leq \frac{TC}{2 \lambda_A}$ is written, with $\omega^2 \leq 0$ also noted.

- With high risk aversion, all portfolios will be same, so tracking error will approach 0. With higher TC, higher the tracking error. Dispersion is directly proportional to TE of each portfolio relative to the composite; the constant of proportionality depends on the number of stocks being managed.

Handwritten diagram illustrating the relationship between dispersion and tracking error. It shows a normal distribution curve with a shaded area under the curve. The formula $\Phi^{-1}(\frac{1}{2}) = 0$ is written. Below this, the formula $\mathbb{E}[r_{max} - r_{min}] = 2 \cdot \Phi^{-1}\left(\left\{\frac{1}{2}\right\}^{1/N}\right) \cdot \omega$ is written, with ω labeled as $\# \text{ stocks}$.

- With changing alpha and risk, it's likely that portfolios will converge. This has been empirically verified.
- To reduce dispersion, once can use dual-benchmark optimization where on top of penalizing active risk w.r.t benchmark we also penalize relative risk to the composite/optimal portfolio. But this will come at a cost of sacrificing higher return.

Handwritten formula for dual-benchmark optimization: $\alpha_p - \lambda_1 \omega_p^2 - \lambda_2 \omega_c^2$.